

$$\text{sub}_{\text{alt}}(\text{stmt}_1; \text{stmt}_2) = \{ \text{stmt}_1; s' \mid s' \in \text{sub}_{\text{alt}}(\text{stmt}_2) \setminus \{\text{exit}\} \} \cup \text{sub}_{\text{alt}}(\text{stmt}_1)$$

$$\begin{aligned} & \text{sub}_{\text{alt}}(\mathbf{do} \ :: \ g_1 \rightarrow \text{stmt}_1 \ \cdots \ :: \ g_n \rightarrow \text{stmt}_n \ \mathbf{od}) \\ &= \{\text{loop-cmd}, \text{exit}\} \cup \bigcup_i \{ \text{loop-cmd}; s' \mid s' \in \text{sub}_{\text{alt}}(\text{stmt}_i) \setminus \{\text{exit}\} \} \end{aligned}$$

$$\frac{\text{stmt}_2 \xrightarrow{g:\alpha} \text{stmt}'_2 \quad \text{stmt}'_2 \neq \text{exit}}{\text{stmt}_1; \text{stmt}_2 \xrightarrow{g:\alpha} \text{stmt}_1; \text{stmt}'_2} \qquad \frac{\text{stmt}_2 \xrightarrow{g:\alpha} \text{exit}}{\text{stmt}_1; \text{stmt}_2 \xrightarrow{g:\alpha} \text{stmt}_1}$$

$$\frac{\text{stmt}_i \xrightarrow{h:\alpha} \text{stmt}'_i \quad \text{stmt}'_i \neq \text{exit}}{\text{loop-cmd} \xrightarrow{(g_i \wedge h):\alpha} \text{loop-cmd}; \text{stmt}'_i}$$